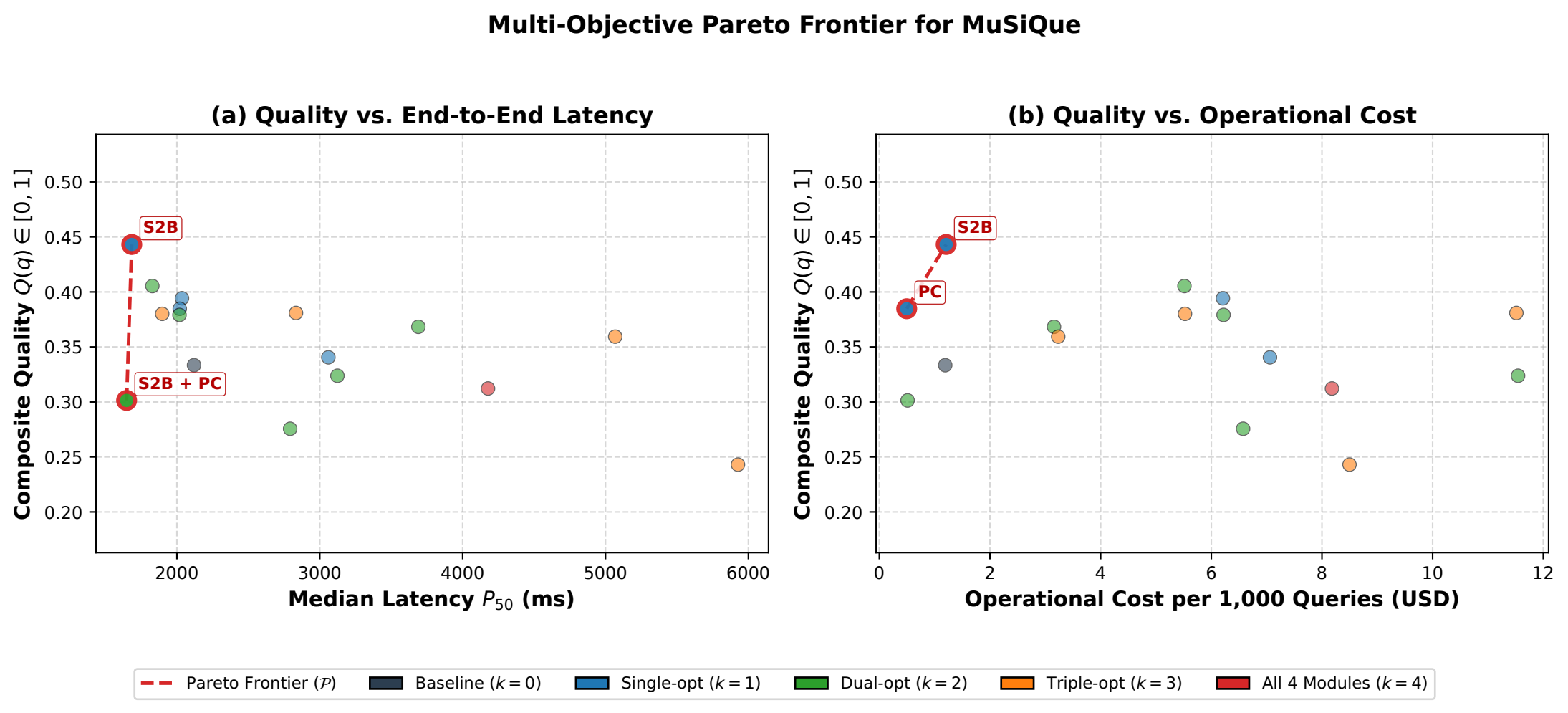
Multi-Objective Pareto Frontier for MuSiQue
(a) Quality vs. End-to-End Latency
(b) Quality vs. Operational Cost
Composite Quality $Q(q) \in [0, 1]$
Median Latency P_{50} (ms)
Operational Cost per 1,000 Queries (USD)
S2B
S2B + PC
PC
0.50
0.45
0.40
0.35
0.30
0.25
0.20
2000
3000
4000
5000
6000
0
2
4
6
8
10
12
Pareto Frontier ($\mathcal{P}$)
Baseline ($k = 0$)
Single-opt ($k = 1$)
Dual-opt ($k = 2$)
Triple-opt ($k = 3$)
All 4 Modules ($k = 4$)